Independence and Conditional Probability

Events A and B are independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently, A and B are independent

if $P(A|B) = P(A)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

(provided $P(B) \neq 0$)

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Conditional Probability: "Unraveling" the Definition

For events A and B,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\Downarrow$$

$$\boxed{P(A \cap B) = P(A|B) \cdot P(B)}$$

Conditional Probability: "Unraveling" the Definition

Similarly,

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)}$$

$$\Downarrow$$

$$\boxed{P(A \cap B) = P(B|A) \cdot P(A)}$$

Conditional Probability: "Unraveling" the Definition

For discrete random variables X and Y

$$P(X=x \mid Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

$$\Downarrow$$

$$P(X=x, Y=y) = P(X=x \mid Y=y) \cdot P(Y=y)$$

Similarly, $P(X=x, Y=y) = P(Y=y \mid X=x) \cdot P(X=x)$

Finding Probabilities by Conditioning

Let X and Y be discrete random variables.

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$$= \sum_y P(X=x \mid Y=y) \cdot P(Y=y)$$

"conditioning on Y"

"Double Conditioning"

Suppose that the goal is to compute

$$P(X=x \mid Y=y)$$

and that it is convenient to do so by conditioning on a third random variable Z.

Don't write $P(X=x \mid Y=y \mid Z=z)$

write $P(X=x \mid Y=y, Z=z)$.

"Double Conditioning"

Goal is to compute: $P(X=x \mid Y=y)$

Before: $P(X=x) = \sum_z P(X=x \mid Z=z) \cdot P(Z=z)$

Now, $P(X=x \mid Y=y)$

$$= \sum_z P(X=x \mid Y=y, Z=z) \cdot P(Z=z \mid Y=y)$$

Proof that $P(X=x \mid Y=y)$

$$= \sum_z P(X=x \mid Y=y, Z=z) \cdot P(Z=z \mid Y=y)$$

The right-hand side is

$$\sum_z \frac{P(X=x, Y=y, Z=z)}{P(\cancel{Y=y, Z=z})} \cdot \frac{P(\cancel{Y=y, Z=z})}{P(Y=y)}$$

$$= \frac{1}{P(Y=y)} \sum_z P(X=x, Y=y, Z=z)$$

$$= \frac{1}{P(Y=y)} \cdot P(X=x, Y=y) = P(X=x \mid Y=y) \checkmark$$